SYSTEMS
MANAGEMENT



From Scale to Security: Evolving GPU Management Standards in 2026

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OCP PLATINUM Member

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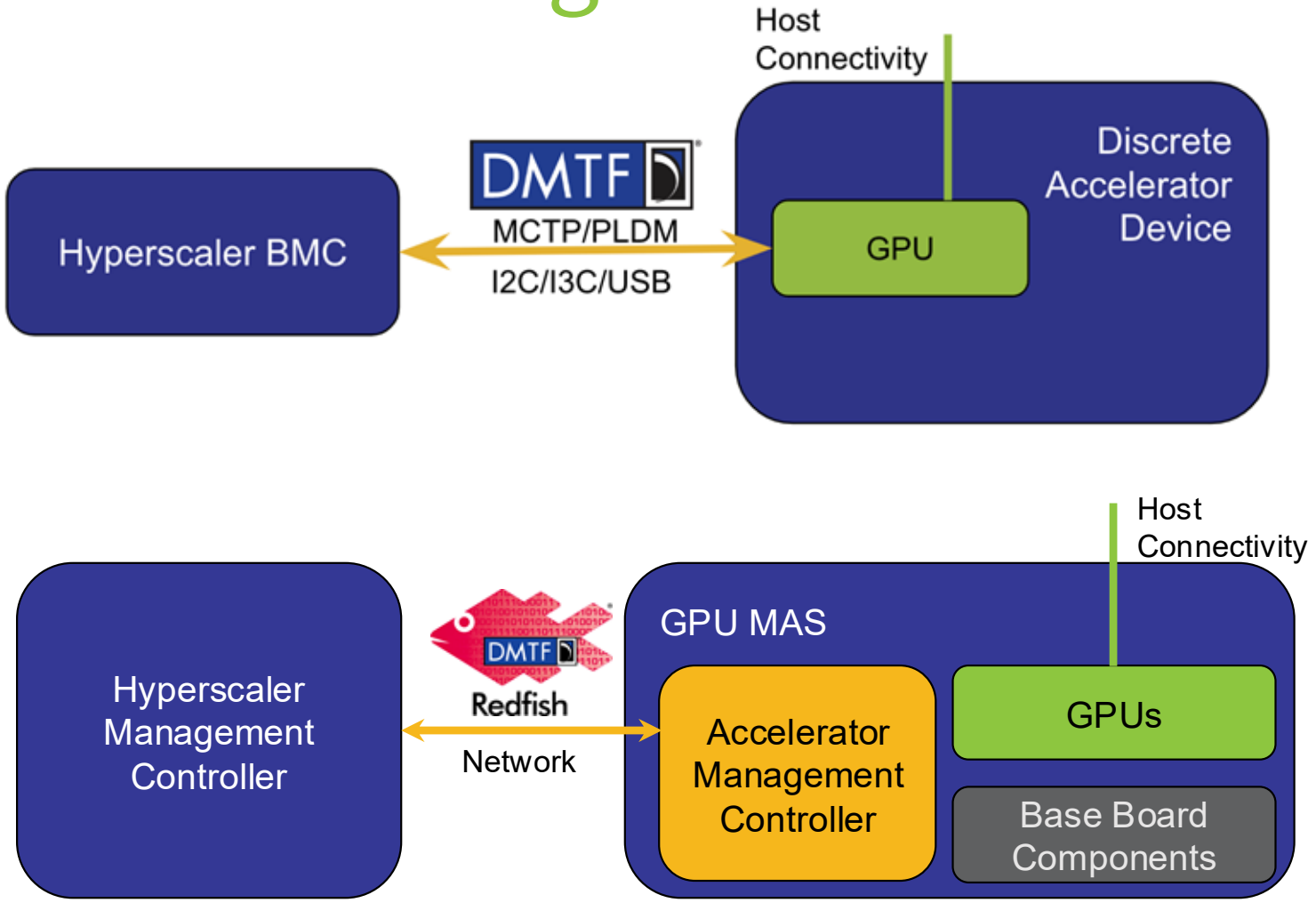
GPU Management Standardization

- Started in 2023…and still going strong
- Goal: Common view of GPU management plane across Hyperscalers and vendors
- Benefits: Seamless integration and deployment of GPU Infrastructure with velocity
- Workstreams:
  - Management Interfaces - 2025 https://www.opencompute.org/documents/ocp-gpu-accelerator-management-interfaces-v1-pdf
  - Firmware Update - 2025 https://www.opencompute.org/documents/ocp-gpu-fw-update-specification-v1-1-pdf
  - RAS - 2025 https://www.opencompute.org/documents/ocp-gpu-and-accelerators-ras-requirements-v1-7-10-23-2025-pdf
  - Compliance Test Tool - https://github.com/opencomputeproject/ocp-diag-ctam

Workstream Overview - Management Interfaces

We define management interface requirements for GPU products in two general areas:

- Discrete Accelerator Devices
  - e.g. PCIe CEM
  - DMTF MCTP, PLDM, SPDM
- Managed Accelerator Subsystems (MAS)
  - e.g. HGX
  - DMTF Redfish



What’s New - Management Interfaces

- To be included in the GPU Management Interfaces 2.0 specification in 2026:
  - GPU Telemetry work that kicked off in 2025 is finalizing and will be included in the GPU Management Interfaces 2.0 specification in 2026.
  - Standardizing a mechanism to trigger desired Redfish events to fully test backend response flows throughout the product lifecycle.
  - Enhancements to the GPU Management Interfaces Redfish Interoperability Profile to include additional properties.

GPU Telemetry Standardization

- Telemetry is indispensable for hyperscale GPU fleet management
- Maintains fleet health: proactive detection of overheating, leaks, abnormal power draw, and impending hardware failures
- Measures fleet performance: utilization, processing speed, memory bandwidth, and latency for bottleneck analysis and capacity planning

The Scale Challenge

- Amortize metadata over larger blocks of telemetry — attach metadata once per block instead of per data point, reducing overhead
- Guarantee minimum requirements for subscription and streaming — services subscribe to real-time or historical streams at a consistent, predictable rate

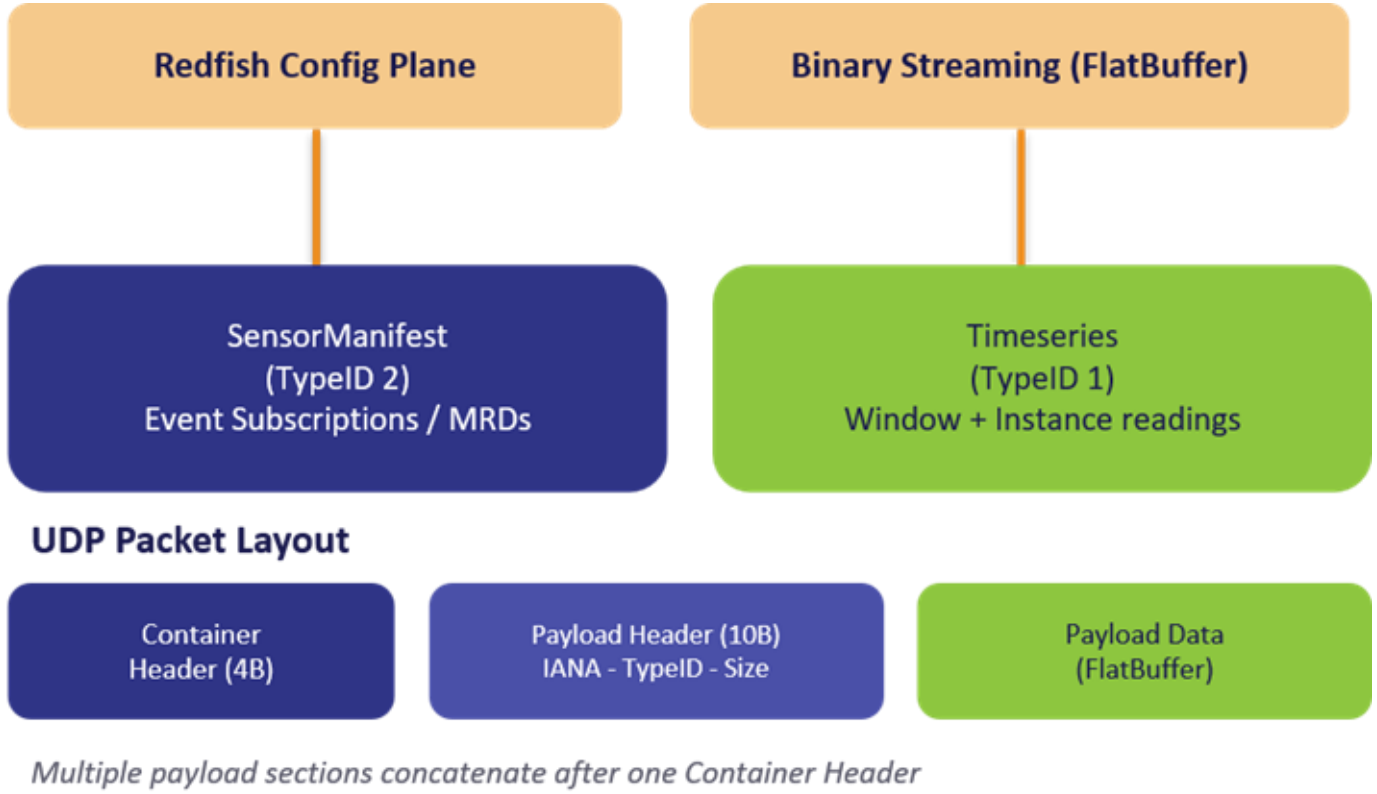
Telemetry Architecture: Config vs. Data

Configuration plane stays on Redfish

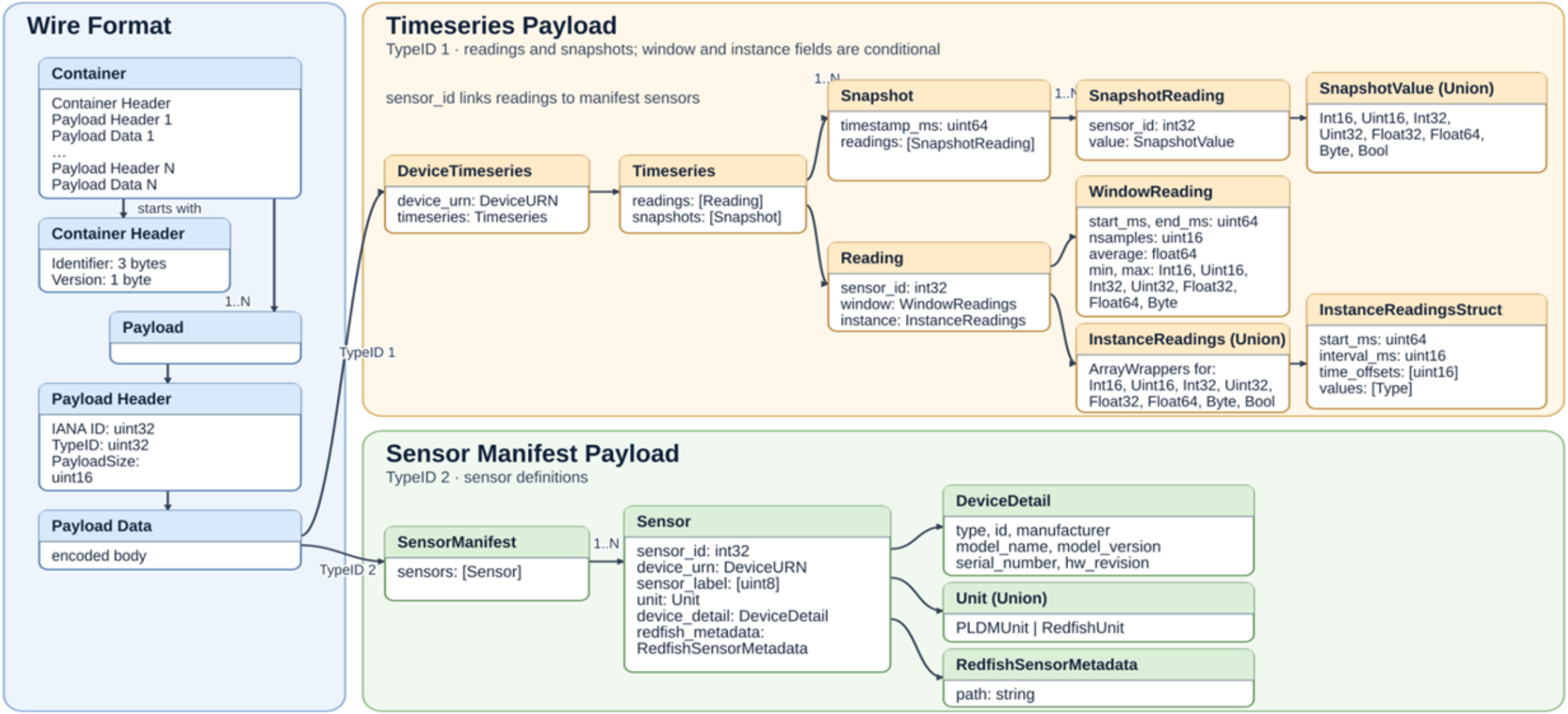
- Event Subscriptions (SSE) and Metric Report Definitions (MRDs) define what streams and filters
- Preserves ecosystem compatibility

Data encoding is via an efficient binary format

- FlatBuffer payloads replace JSON for high-volume streaming
- Addresses efficiency & scalability limits at hyperscale



Sensor Manifest & Timeseries: Correlated by sensor_id

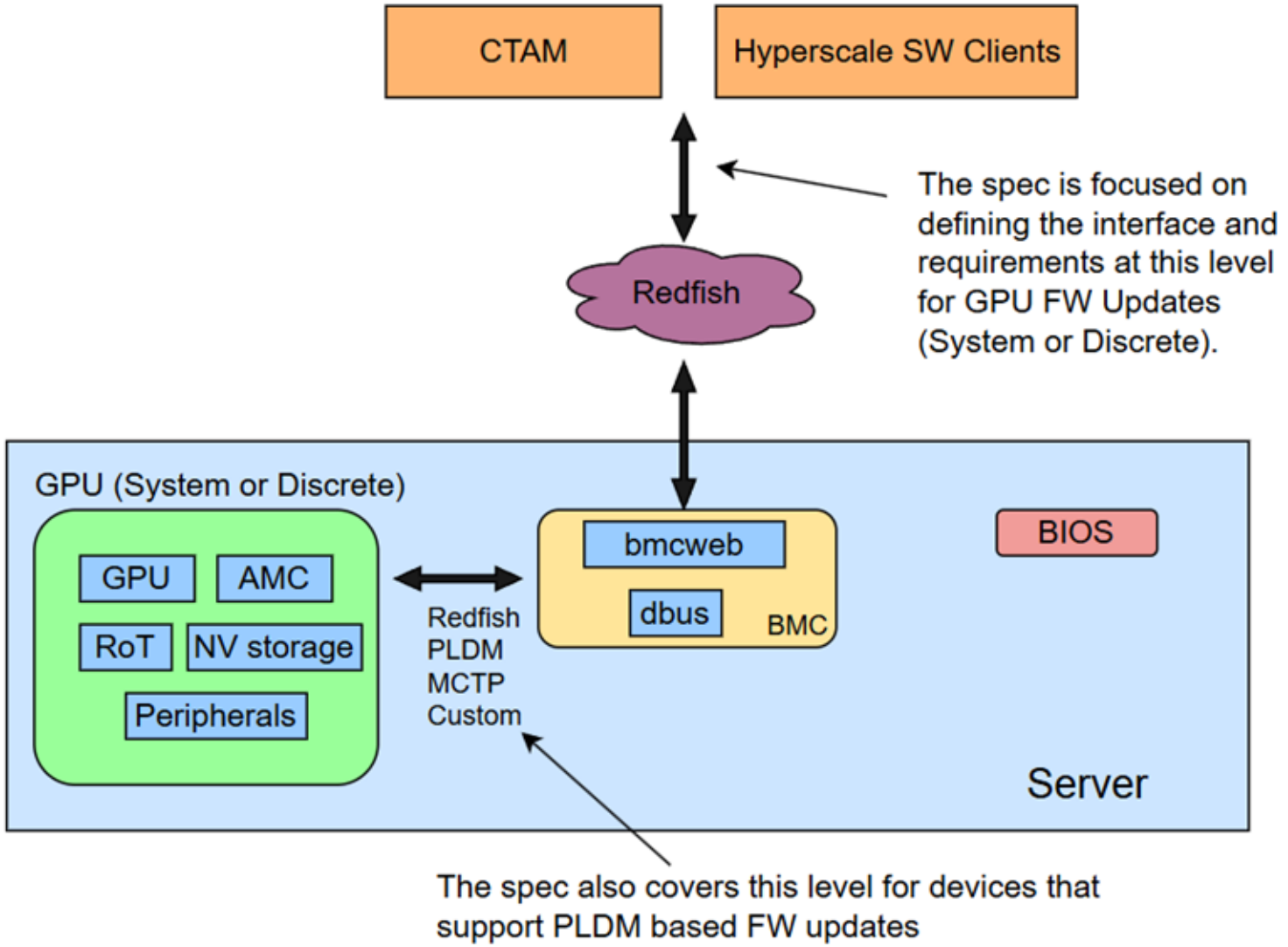


Redfish Event Discovery and Simulation

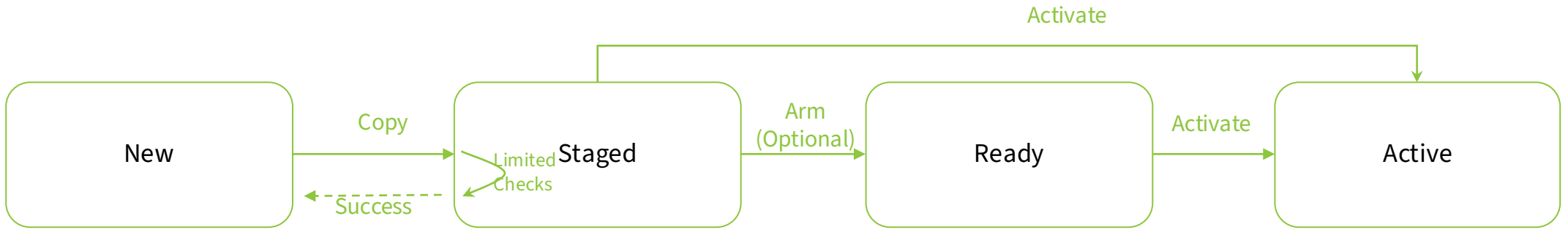
- Event Discovery
  - Standardize an OCP mechanism to discover every supported Redfish event and its properties.
  - Working with DMTF on ways to expose supported message IDs. Till then we have a standard schema for suppliers to communicate this with Hyperscalers.
- Event Simulation
  - Enables hyperscalers to develop and validate fleet event-handling workflows before deployment, including for events that cannot otherwise be injected.
  - DMTF already supports
    - /redfish/v1/EventService/Actions/EventService.SubmitTestEvent

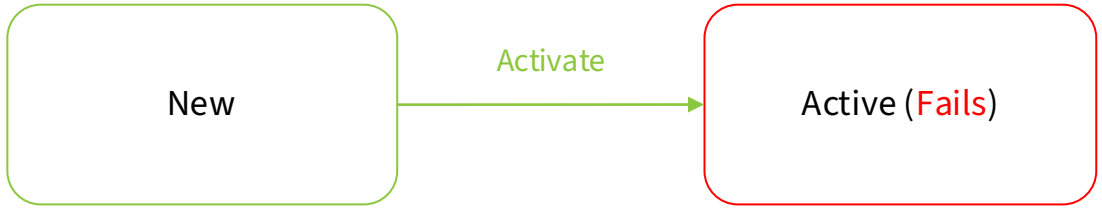
FW Update Workstream Overview

- FW Update is one of the most critical operations on GPUs to maintain
  - Stability, Security, Performance
- Includes all mutable components
- Define and formalize Capabilities required to deploy and operate GPU Systems at scale
- Define interfaces to support the capabilities
- Version 1.1 released in Oct'2025
- Version 1.9 is targeted for Oct'2026
  - Fast Fail with eligibility checks
  - Security Feature Set with controls and versioning support



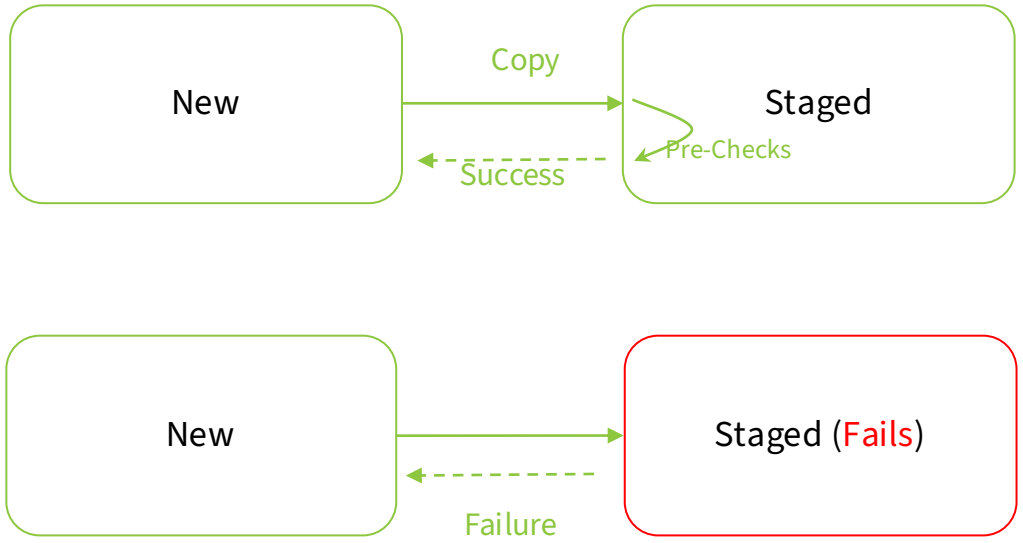
Lack of Fast Fail in FW Update - Problem





- Checks during Copy/Staging is limited, primarily focused on Package integrity
- Pre-checks lack definitions and are implementation specific
- Results in disruption and toil

Fast Fail in FW Update - Solution



- Perform rigorous pre-checks during stage/copy
- Primary goal of pre-check is to ensure that all the targeted devices have met all the criteria for activating the FW
- Early failure indication avoids expensive disruptions

Fast Fail Interface

- Optional Parameter in Multipart “PreUpdateValidation”

```
"Oem": {
        "OpenCompute_SystemMgmt":{
            "@odata.type": "#<OpenCompute_SystemMgmt>UpdateService.v1_0_0.<OpenCompute_SystemMgmt>UpdateService",
            "PreUpdateValidation": true
        }
    }
}
```

- If specified and implemented by the vendor, fast fail checks will be performed
- The parameter applies to full bundle updates
    - Behavior when specifying device updates using “Target” is implementation specific

Fast Fail Errors

Error Type	Description
Pre-Update Validation Failed	One or more targets failed pre-checks
Missing Component in Package	A component FW image is missing in the package
Device missing	A device is absent or cannot be enumerated
Device Powered off	A device is powered off or in a power state that will prevent its FW update
Device in recovery	A device is in recovery state and is not updateable
Management Interface unavailable	A device's management interface (USB, I2C etc.) is down

Firmware Update Security

Context:

- Minimum Security Version (MinSVN) in the RoT sets downgrade version threshold
- A FW package contains its SVN which is used for the threshold
- A FW package contains the desired MinSVN
- MinSVN is updated during FW Update process

Problem:

- FW Rollouts in the fleet locks in MinSVN
- If an issue is found later, rollback becomes extremely challenging

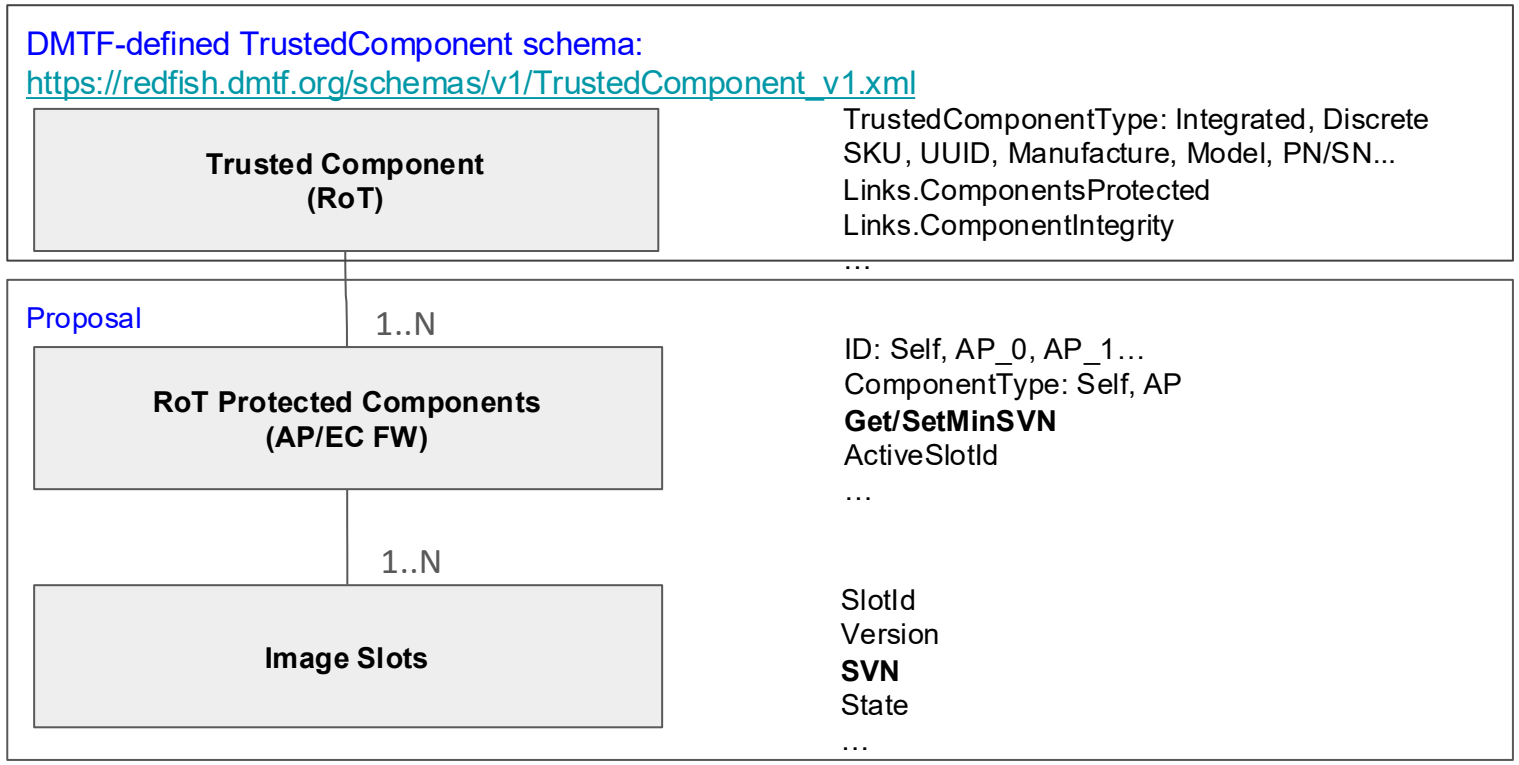
Solution:

- Decouple MinSVN “Burn” from FW Update
- “Burn” MinSVN via a separate Update when ready

FW Update Security Proposal

- Able to Query Security Version Number (SVN) of a FW instance (Staged or Active)
- Able to Query MinSVN of the Device RoT
- Able to securely update MinSVN of the Device RoT

Min SVN Interface proposal



Summary/Call to Action

- Summary
  - GPU Telemetry & event testing standardization
  - Progress towards toil-less and robust FW Update at scale
- Future Plans
  - Adopt Redfish standard for copy/Arm support
  - Solidify PLDM bundle requirements
  - Seamless Updates
  - Redfish Metric ID and String ID standardization
  - Continue RIP enhancements
- Provide feedback to the community via hyperscale-gpu@ocproject.net

Thank You!

